Island Economy Exercise

Macroeconomics
Scarcity and Choice

Part A: Scarcity and Trade-Offs (Round 1)

Decision Example (10 tokens):

- Fish Produced: 6
- Shelters Produced: 2

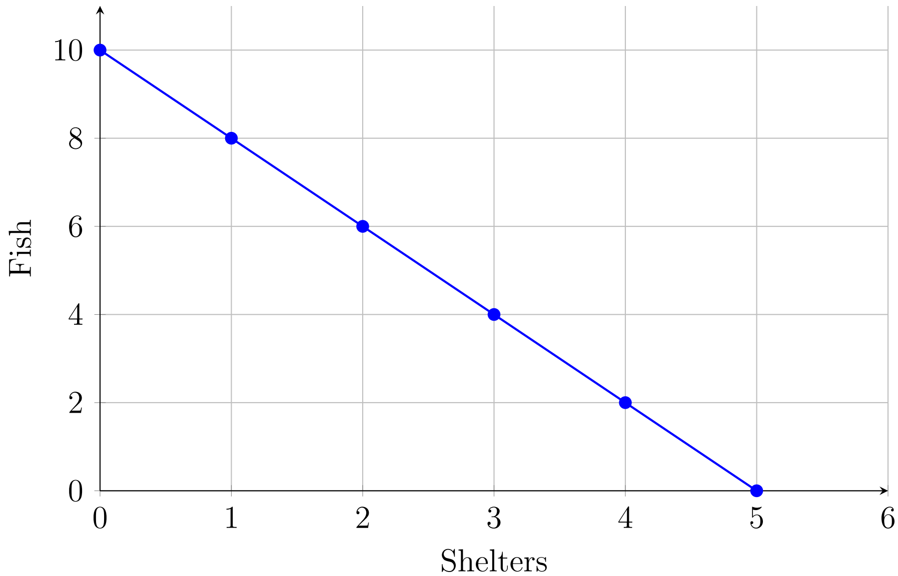
Sample Answers:

1. The trade-off was between food (fish) and protection (shelters). More fish means fewer shelters, and vice versa.
2. To produce 1 more shelter (cost = 2 tokens), the group must give up 2 fish. Opportunity cost of 1 shelter = 2 fish.

Part B: Production Possibilities Frontier (PPF)

PPF Points with 10 Tokens:

$$\begin{gathered}
0 \text{ Shelters} \rightarrow 10 \text{ Fish} \\
1 \text{ Shelter} \rightarrow 8 \text{ Fish} \\
2 \text{ Shelters} \rightarrow 6 \text{ Fish} \\
3 \text{ Shelters} \rightarrow 4 \text{ Fish} \\
4 \text{ Shelters} \rightarrow 2 \text{ Fish} \\
5 \text{ Shelters} \rightarrow 0 \text{ Fish}
\end{gathered}$$



Note: The PPF is linear in this setup because opportunity cost is constant (1 shelter = 2 fish).

Part C: Specialization and Trade (Round 2)

Example:

- Group A (Fishing Advantage): 10 tokens → 20 Fish
- Group B (Shelter Advantage): 10 tokens → 10 Shelters

If they specialize and trade: - Suppose Group A trades 10 fish for 4 shelters with Group B. - Final Bundles: - Group A: 10 Fish + 4 Shelters - Group B: 10 Fish + 6 Shelters
Sample Answers:

1. Specialization increased total production: 20 fish + 10 shelters vs. mixed production with less output.
2. Both groups are better off after trade compared to autarky (no trade).

Part D: Economic Shocks (Round 3)

1. If a storm destroys half of all shelters, shelters become more valuable relative to fish. The opportunity cost of a shelter rises.
2. A new fishing net doubling fishing productivity shifts the PPF outward in the fish direction. Now 1 token = 2 fish, so the maximum is 20 fish (with 0 shelters). The economy can consume more fish for any given number of shelters.

Reflection Questions – Suggested Responses

1. Scarcity is shown because groups cannot produce unlimited amounts of both goods with only 10 tokens.
2. The PPF shows opportunity costs: the slope represents how many fish must be given up for each shelter.
3. Countries specialize and trade for the same reason as the groups — comparative advantage increases total production and allows everyone to be better off.
4. When resources or technology change, the PPF shifts outward (growth) or inward (loss), changing the economy's possible choices.